Practical Test: Prompt Engineering (Beginner Level)

Duration: 45–60 minutes
Tool Allowed: Any LLM (ChatGPT, Gemini, Claude, etc.)

Section A: Prompt Writing (10 Marks)

Question 1

Write a **clear and structured prompt** that asks an AI model to:

> Explain **Machine Learning** to a **class 10 student** using **simple language and real-life examples**.

Expected Skills Tested

- Clear instruction
- Audience specification
- Output clarity

Answer 1:

Explain the concept of Machine Learning to a Class 10 student.
Use simple language, avoid technical jargon, and include 2–3 real-life examples such as recommendation systems, spam email detection, or voice assistants.
Structure the explanation in the following format:

1. Definition of Machine Learning
2. How it works (simple explanation)
3. Real-life examples
4. Why it is important

Question 2

Write a prompt that asks an AI model to generate:

- **5 quiz questions**
- Topic: **Artificial Intelligence**
- Difficulty: **Easy**
- Include **answers**

Answer 2:

Generate **5 quiz questions** about **Artificial Intelligence** for beginners.
Difficulty level should be **easy**.

Provide the output in the following format:

Question 1
Options: A, B, C, D
Correct Answer

Question 2
Options: A, B, C, D
Correct Answer

Ensure the questions are **simple and suitable for beginners**.

Section B: Prompt Improvement (10 Marks)

Below is a **poor prompt**:

"Tell me about AI."

Task:

Rewrite this prompt to make it **more effective and specific**.

Your improved prompt must include:

- Target audience
- Output format
- Examples or explanation

Answer:

Explain **Artificial Intelligence** to **high school students** in simple language.

Structure the answer in the following format:

1. Definition of Artificial Intelligence
2. How AI works (simple explanation)
3. 3 real-life examples of AI (e.g., voice assistants, recommendation systems, self-driving cars)
4. Advantages of AI

Limit the explanation to **150–200 words** and use **easy-to-understand language**.

Section C: Role-based Prompting (10 Marks)

Write a prompt that asks the AI to:

- Act as a **Data Science teacher**
- Explain **Cosine Similarity**
- Include:
    - Formula
    - Simple example
    - Python code snippet

Answer:

Act as a Data Science teacher and explain the concept of Cosine Similarity.

Your explanation should include:

1. A simple definition
2. The Cosine Similarity formula
3. A simple example using vectors
4. A Python code snippet that calculates cosine similarity between two vectors.

Use clear language suitable for beginners learning data science.

Section D: Structured Output Prompt (10 Marks)

Write a prompt that asks the AI to generate a **table** containing:

Algorithm	Use Case	Example

for **5 Machine Learning algorithms**.

Answer:

Generate a **table** containing information about **5 Machine Learning algorithms**.

The table should include the following columns:

| Algorithm | Use Case | Example |

Ensure the explanations are **short and clear**, and suitable for beginners studying machine learning.

Section E: Prompt Debugging (10 Marks)

An AI produced a **very long and complex answer** when asked a question.

Write a prompt that instructs the AI to:

- Give the answer in **5 bullet points**
- Use **simple language**
- Limit response to **100 words**

Answer:

Answer the question in a **clear and concise way** using the following constraints:

• Provide the answer in **exactly 5 bullet points**
• Use **simple and easy language**
• Keep the total response **under 100 words**
• Avoid unnecessary details

Bonus Question (Optional – 5 Marks)

Write a prompt that asks an AI to generate a **Python program that performs sentiment analysis on text**.

Answer:

Write a **Python program** that performs **sentiment analysis on text**.

Requirements:

1. Use a Python library such as **TextBlob or NLTK**

2. Accept a text sentence as input

3. Determine whether the sentiment is **positive, negative, or neutral**

4. Print the sentiment result

Also include comments explaining each part of the code.

Evaluation Rubric

Criteria	Marks
Clarity of prompt	10
Use of constraints	10
Structured prompting	10
Role-based prompting	10
Output formatting	10